INSAF LASHARI

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EDUCATION

University of Waterloo Waterloo, ON

Candidate for Bachelor of Applied Science of Computer Engineering, Honours *Present – May 2028*

EXPERIENCE

Electrical Engineering Intern Sep 2024 – Dec 2024

Greer Galloway Group Inc *Belleville, ON*

- Led an audit to update single-line diagrams for three water treatment plants and two wastewater treatment plants, ensuring accuracy and up-to-date documentation of existing conditions.
- Conducted on-site inspections alongside superintendent to assess existing power distribution, control panels and equipment layouts.
- Verified and documented electrical loads, transformer ratings, and breaker configurations, ensuring proper system documentation for future maintenance and upgrades.
- Created detailed markups and redlines for AutoCAD technicians, ensuring precise updates to electrical drawings in alignment with industry standards and regulatory requirements.
- Conducted on-site assessments for a building renovation project, reviewing existing power, data, lighting, and emergency systems.

Engineering Intern Jan 2024 – Apr 2024

Greer Galloway Group Inc *Belleville, ON*

- Managed 7 major electrical projects and 1 major Request for Supplier Qualifications (RFSQ), assisted with writing tender packages, proposals, reports and conducted on-site construction visits.
- Collaborated on electrical designs for schools, residential projects and hospitals focusing on power distribution, data systems, emergency lighting, and fire alarms.
- Reviewed technical drawings to ensure compliance with the Ontario Building Code (OBC) and Ontario Electrical Safety Code (OESC).
- Designed interior and exterior lighting layouts using AGI32 for diverse applications, including streetlights, parking lots, residential buildings, schools, and apartment complex, optimizing illumination levels, energy efficiency, and compliance with RP8 standards.

PROJECTS

StudyBuddy - Student Productivity App | *Flutter, Dart* May 2025 – Present

* Developed a Pomodoro timer and task manager to enhance focus, time management, and organization for students.
* Built a cross-platform Android and Web app with smooth performance and consistent UI using Flutter.
* Designed a minimalist, intuitive interface with distraction-free study sessions, task tracking, and swipe-to-delete functionality.
* Planned enhancements: game-level system, analytics tracking, and Google Play Store deployment.

Choose Your Own Adventure - AI-Powered Storytelling | *Python (FastAPI), React* May 2025 – Sep 2025

* Built a branching narrative web app with FastAPI backend and React frontend for dynamic storytelling.
* Designed recursive backend architecture with relational database to manage complex story trees and user sessions efficiently.
* Integrated OpenAI GPT-4 to generate on-demand story content, demonstrating AI/NLP integration skills.
* Implemented RESTful APIs with CORS and asynchronous jobs for smooth frontend-backend communication.
* Delivered a responsive, user-friendly UI with state management, loading indicators, error handling, and intuitive interactions.

AWS Math Web Application | *AWS Cloud Practitioner Essentials* May 2025 – Sep 2025

* Developed a serverless web application using AWS Amplify, API Gateway, Lambda, IAM, and DynamoDB to compute mathematical operations and persist results via a secure REST API.
* Demonstrated cloud-native development by integrating multiple AWS services into a functional application.
* Completed AWS Cloud Practitioner Essentials to gain foundational knowledge of AWS services, architecture, and best practices.

TempCtrl - FPGA HVAC Energy Monitoring System | *VHDL, FPGA* May 2024 – Aug 2024

* Designed a digital HVAC control system on FPGA using VHDL, integrating temperature comparators, window/door sensors, and automated furnace/AC/blower logic.
* Developed modular digital hardware with simulation-driven verification, including vacation mode logic and 7-segment display interfaces for real-time monitoring.

VeriLight - FPGA Traffic Light Controller | *VHDL, FPGA* May 2024 – Aug 2024

* Built a traffic light control system on FPGA in VHDL using a finite state machine (FSM) to coordinate traffic signals, pedestrian requests, and timed sequences.
* Implemented debouncers, synchronizers, holding registers, and 7-segment display drivers, validating functionality through simulation and waveform analysis.

TECHNICAL SKILLS

Languages: C++, C, Python, JavaScript (ES6+), HTML5, CSS3, VHDL, Kotlin, MATLAB, RISC-V, ARM
Frameworks: React.js (with JSX), Vue.js, Flutter, FastAPI, Node.js
Developer Tools: Git, Vite, Visual Studio Code, Adobe Creative Cloud, AGi32, COMSOL, Microsoft Office, Google Suite, AutoCAD
Cloud and Databases: AWS (Amplify, Lambda, IAM, API Gateway, DynamoDB), MySQL

CERTIFICATIONS

Tech Stewardship Practice Program Jan 2024 – Mar 2023
AWS Cloud Practitioner Essentials Jun 2025 – Aug 2025